

C#.NET @ Sunbeam Infotech

Trainer: Nilesh Ghule



Day 02 - .Net

① .NET versions

② C# lang versions

- latest C# 14 for .Net 10.

③ 01_HelloWorld

- ✓ - .Net Core Hello
- Top level stmts

④ 02_Namespaces

- Logical containers
- default namespace
- Split namespace
- File scoped ns
- Implicit usings for common ns.
- Namespace alias
- global usings

* multiple entry points.

⑤ 03_Data Types.

- Parse()
- Convert
- checked/
unchecked

⑥ 04_Struct Enums

- Struct: props, ctor x
- obj initializers
- collection init
- class vs struct

⑦ 05_boxing

- $\text{int} \rightleftharpoons \text{object}$
- boxing needed in non-gen collections, but not in gen col.

⑧ 06_methods

- pass by val/ref
- ref & out keyword
- local methods

⑨ 07_inheritance

- core concepts
- base ctor call
- new method (hiding)
- obj slicing
- is keyword
- as keyword
- virtual methods
- new virtual

⑩ 08_inheritance

- inheritance hierarchy
- util methods - static class
- abstract method
- abstract class
- sealed override
- sealed class
- array of objs



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-  Console App (.NET Framework) C#
-  Class Library (.NET Framework) C#
-  Blank Solution
-  Windows Forms App C#

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Windows

Console



Console App

A project for creating a command-line application that can run on .NET on Windows, Linux and macOS

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Other

Solution name

Day02Soln

Location

D:\Nilesh\Feb25\DAC\dotnet\day02

Solution will be created in "D:\Nilesh\Feb25\DAC\dotnet\day02\Day02Soln\"

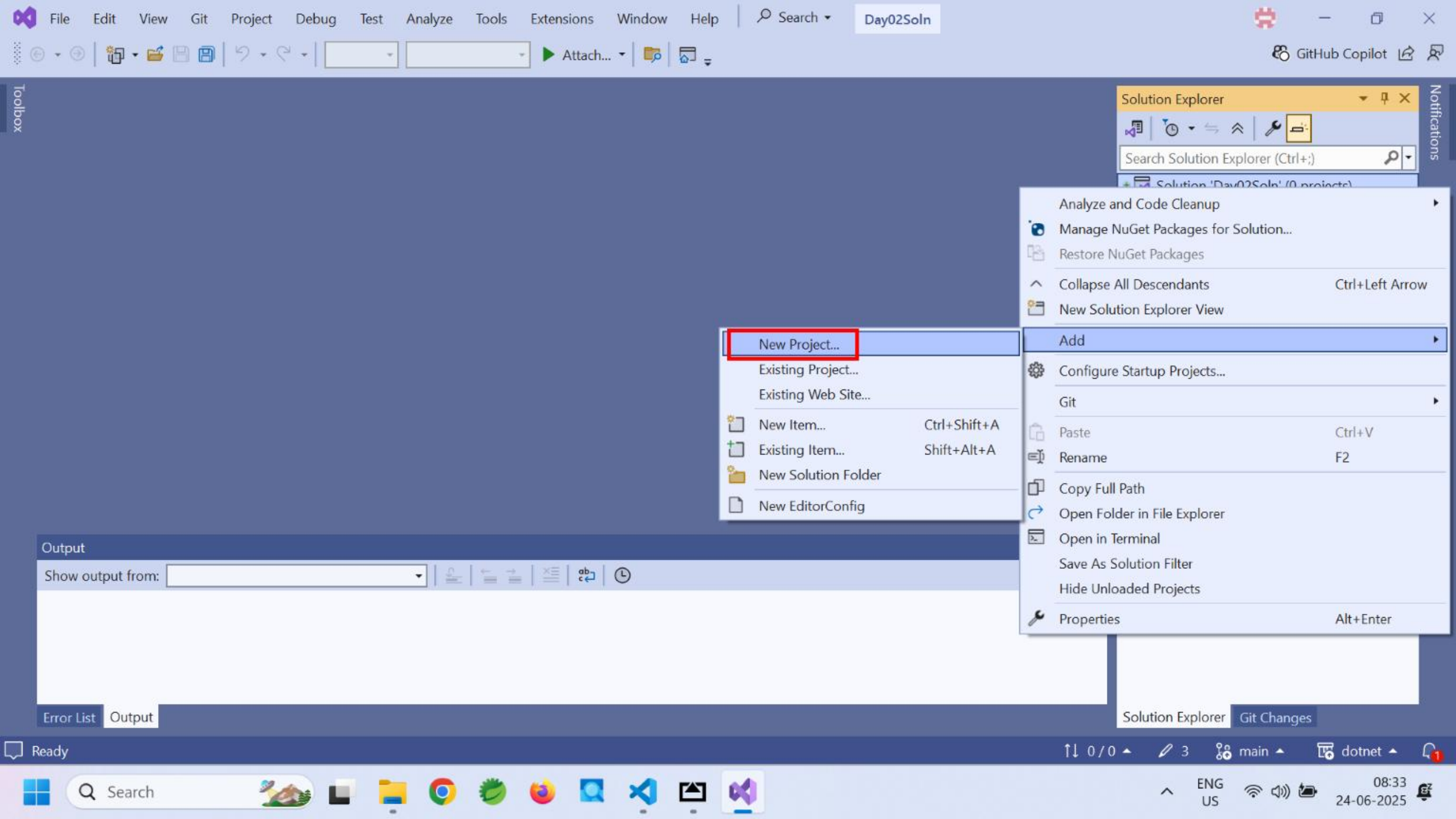
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Configure your new project

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Project name

01_HelloWorld

Location

D:\Nilesh\Feb25\DAC\dotnet\day02\Day02Soln

Project will be created in "D:\Nilesh\Feb25\DAC\dotnet\day02\Day02Soln\01_HelloWorld"

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Additional information

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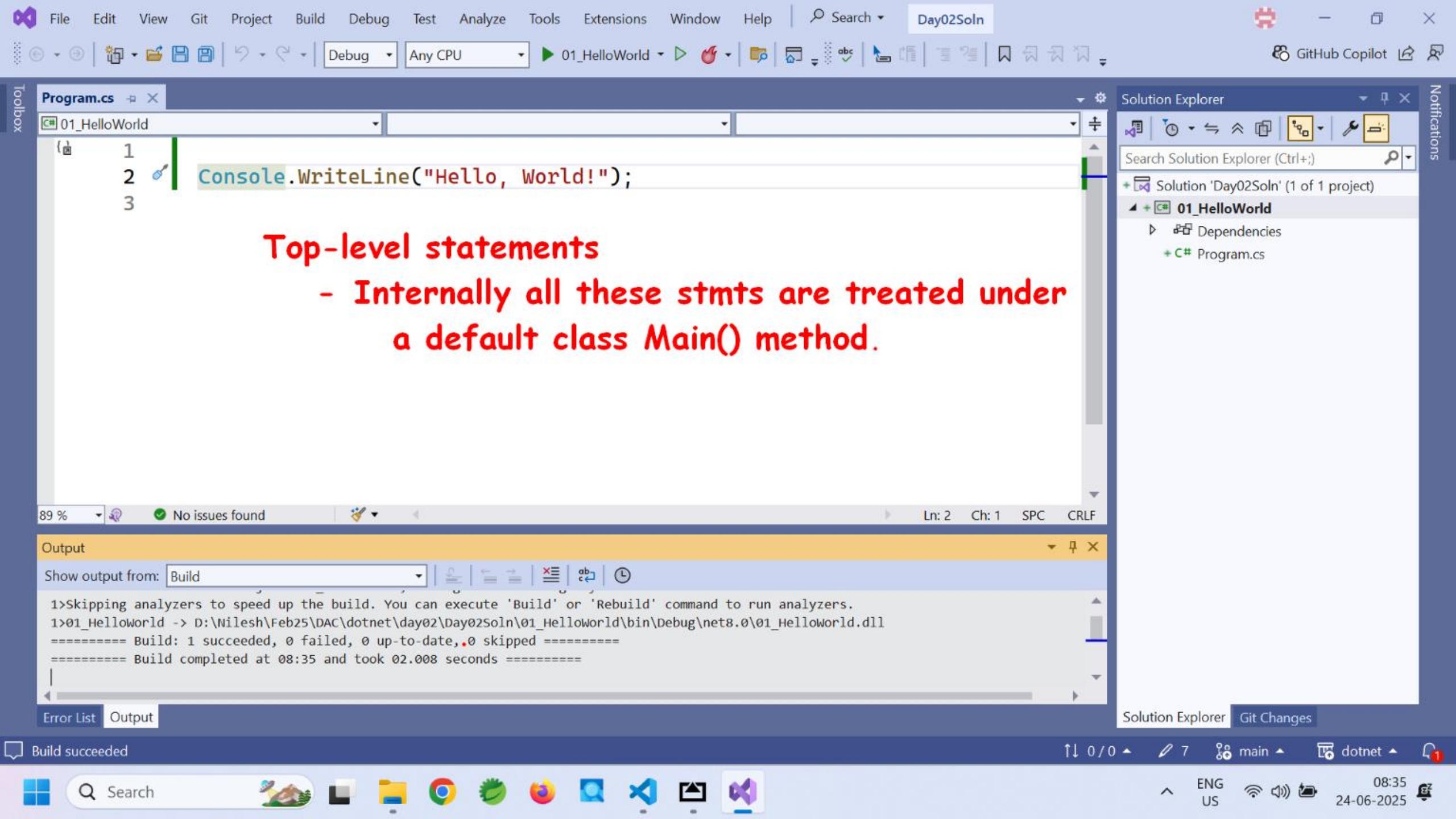


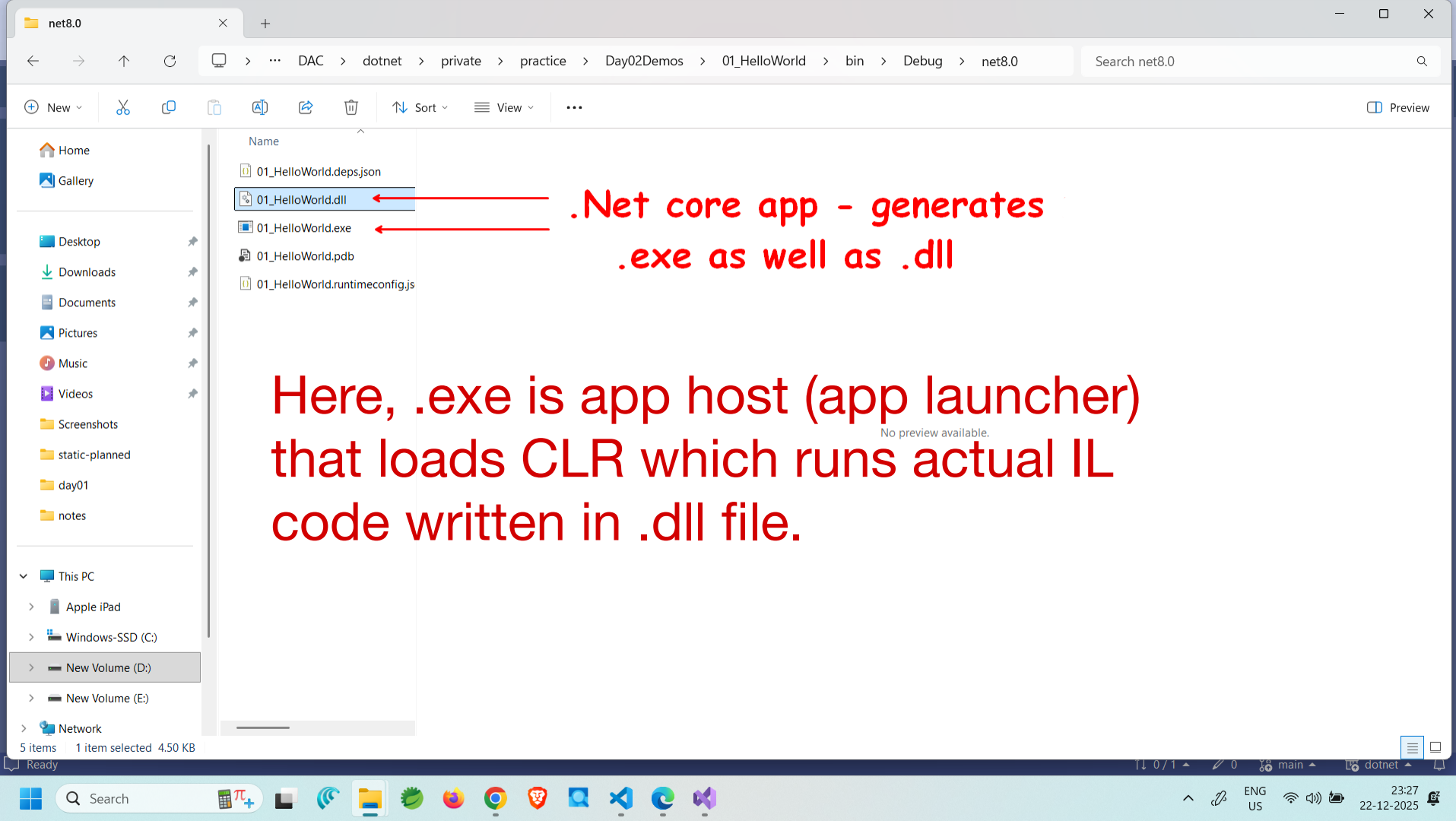
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C#

 Class Library (.NET Framework)

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C#

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Output

Show output from

```
1>Skipping an
1>01_Hellowor
===== BU
===== BU
```

Error List Output

This item does not s



Search



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US



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dotnet

Additional information

Console App

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Create

Output

Show output from

```
1>Skipping an
1>01_Hellowor
===== Bu
===== Bu
```

Error List Output

This item does not s



Search



ENG
US



08:38
24-06-2025



FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchDay02Soln

01_HelloWorld01_HelloWorld

Program.cs

02_Namespaces02_Namespaces.ProgramMain(string[] args)

```
12  /*
13  Namespaces are logical container.
14  - Code organization -- Logical hierarchial arrangement
15  */
```

System.Data.dll

↓

Top Namespace

Nested Namespace

namespace System.Data;

System.Data.SqlClient

SqlConnection
SqlCommand
...

System.Data.OleDb

OleDbConnection
OleDbCommand
...

System.Data.Odbc

OdbcConnection
OdbcCommand
...

Typically namespaces are mapped to assemblies.

- One Asm (.dll) may have multiple namespaces.

- One Asm may have one namespace.

89 %No issues found

Ln: 14Ch: 59SPCCRLF

Error ListOutput

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'Day02Soln' (2 of 2 projects)

01_HelloWorld

Dependencies

Program.cs

02_Namespaces

Dependencies

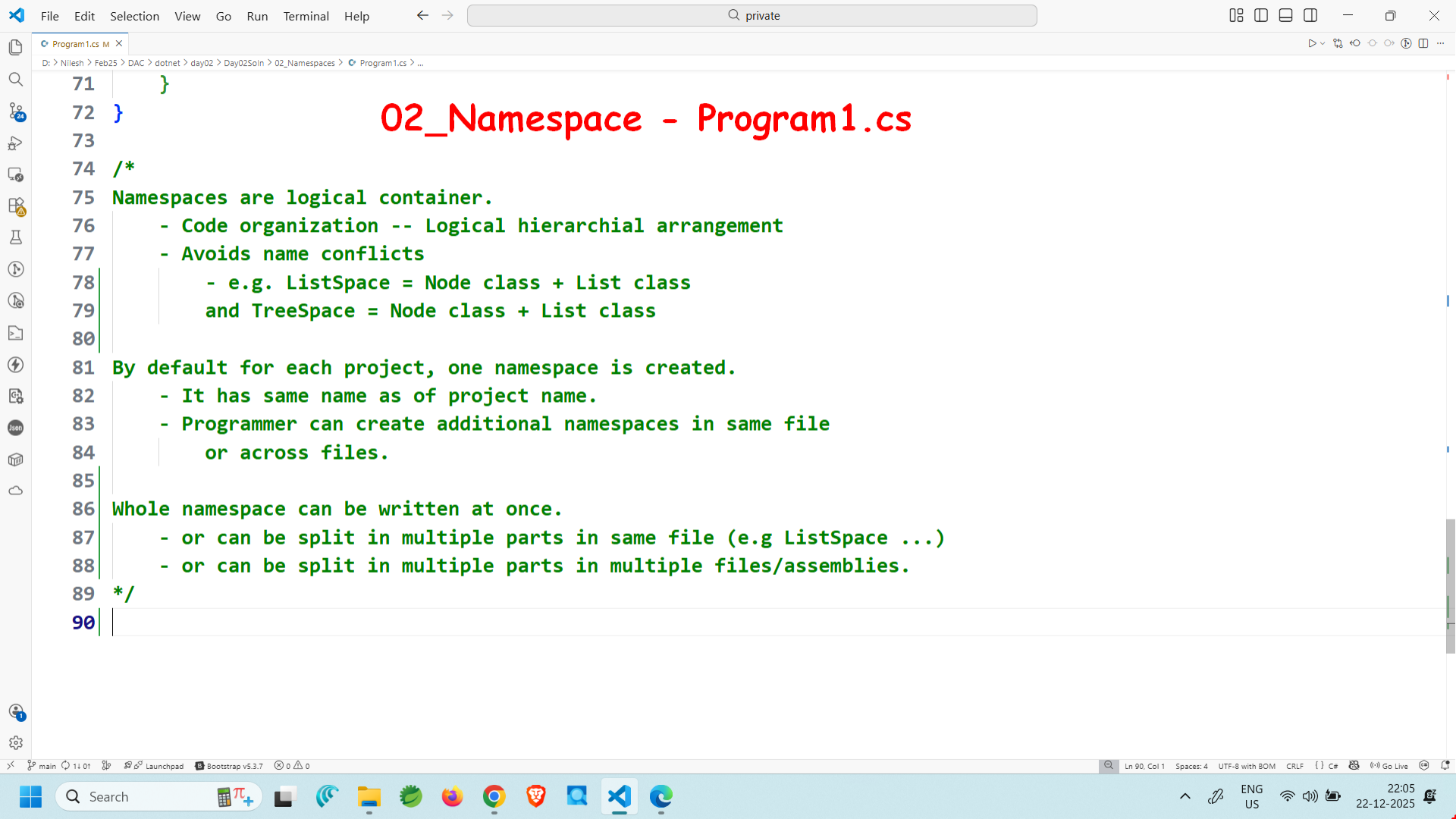
Program.cs

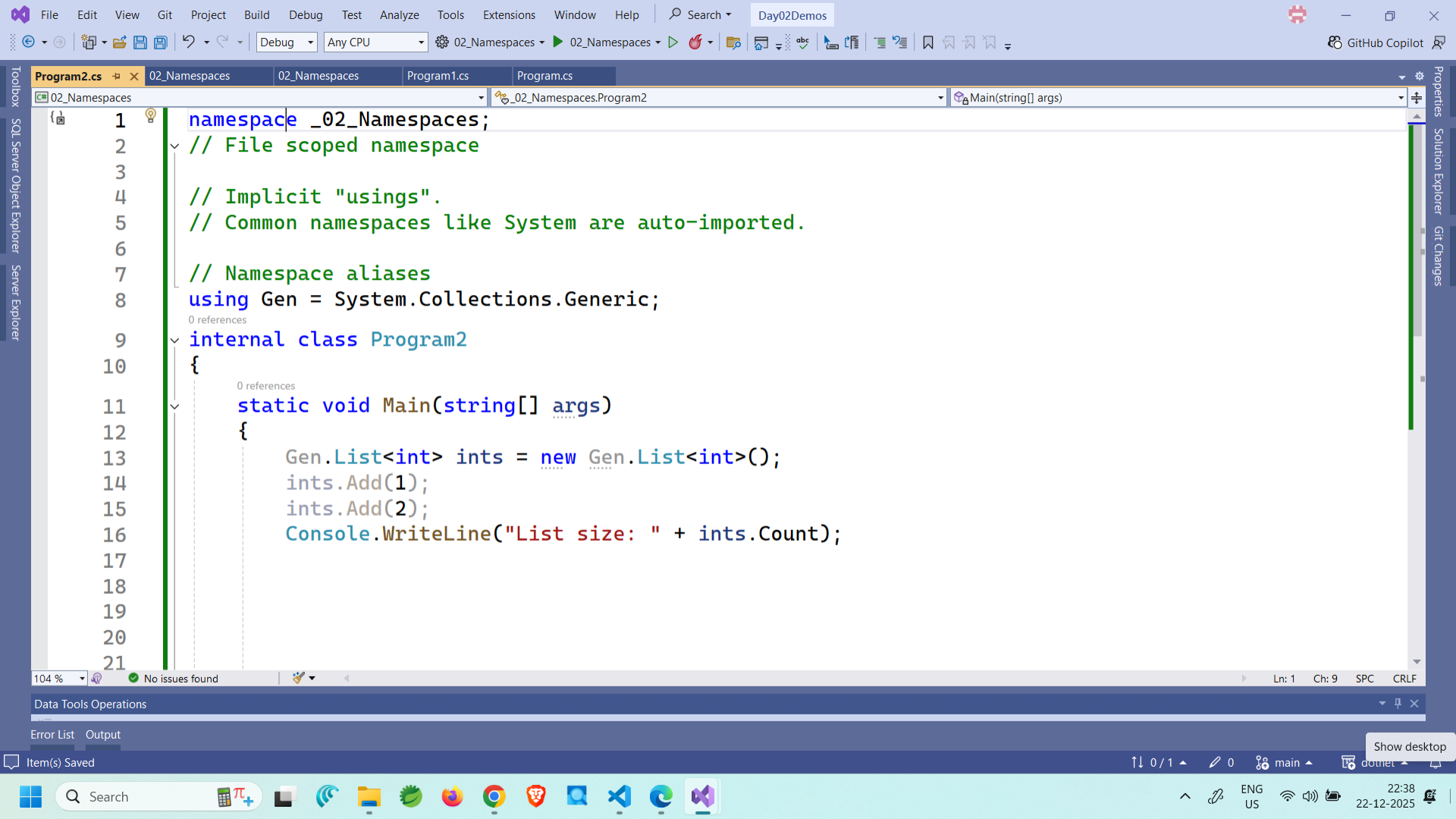
Item(s) Saved

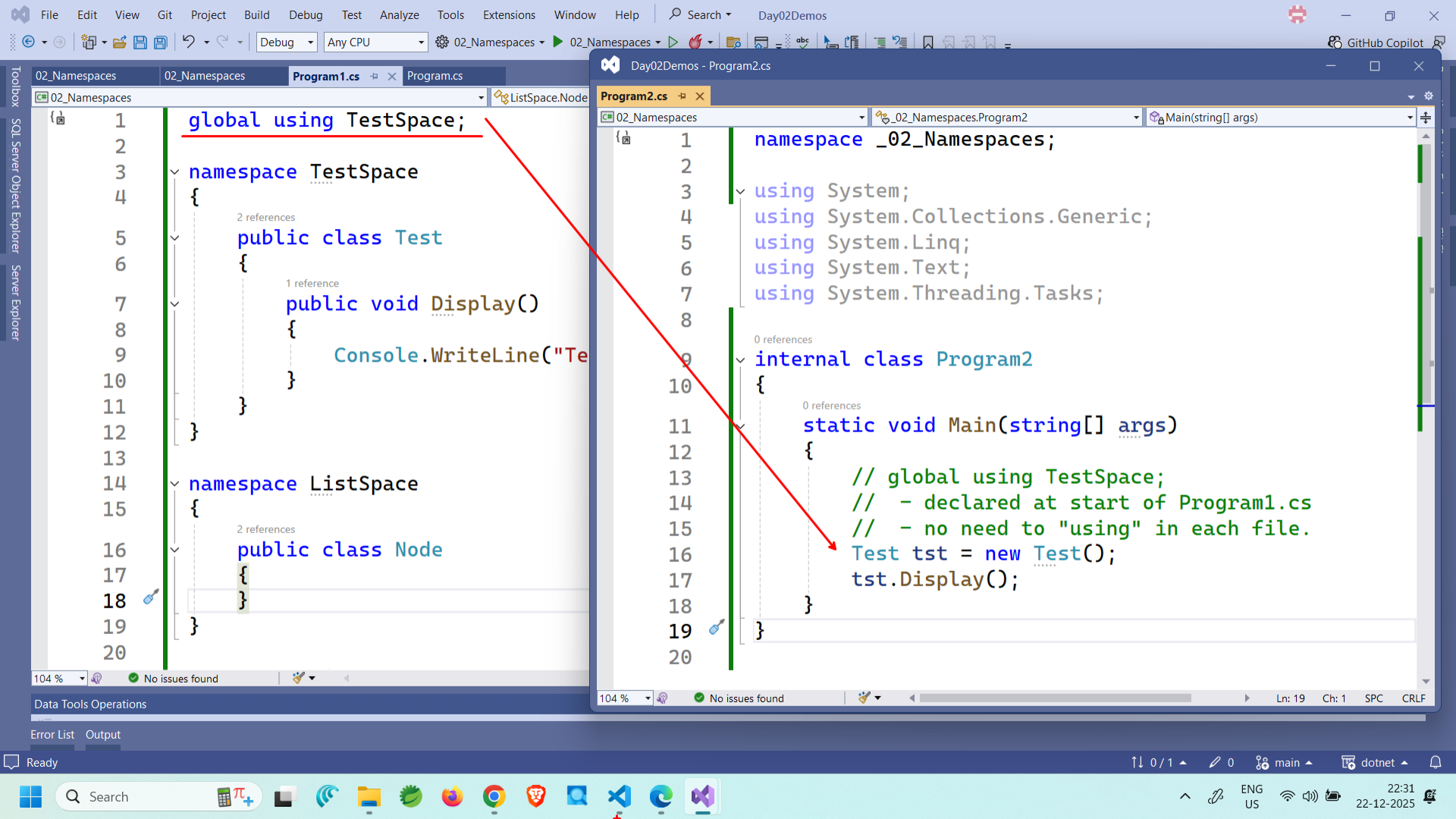
Search

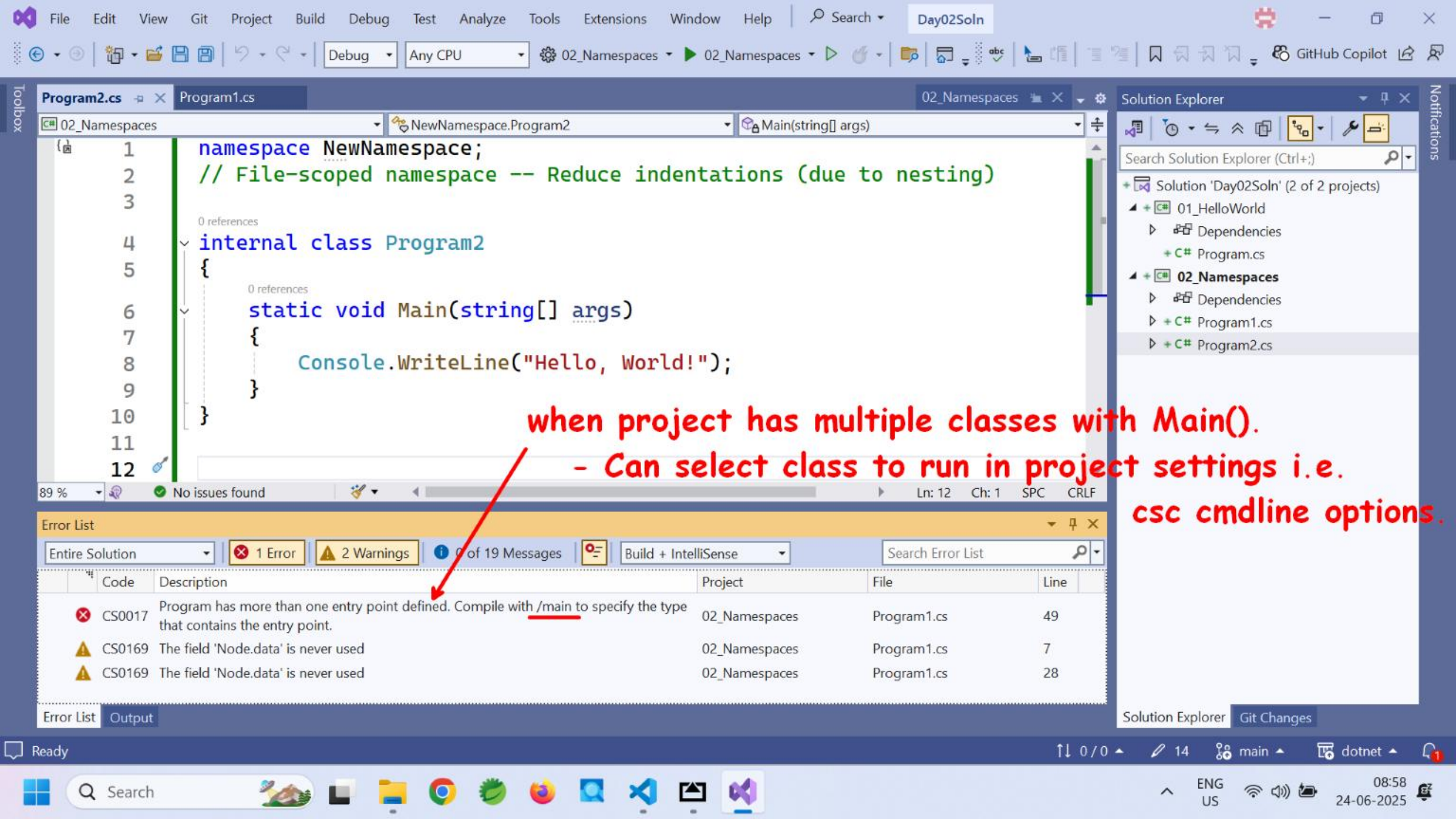
ENG US

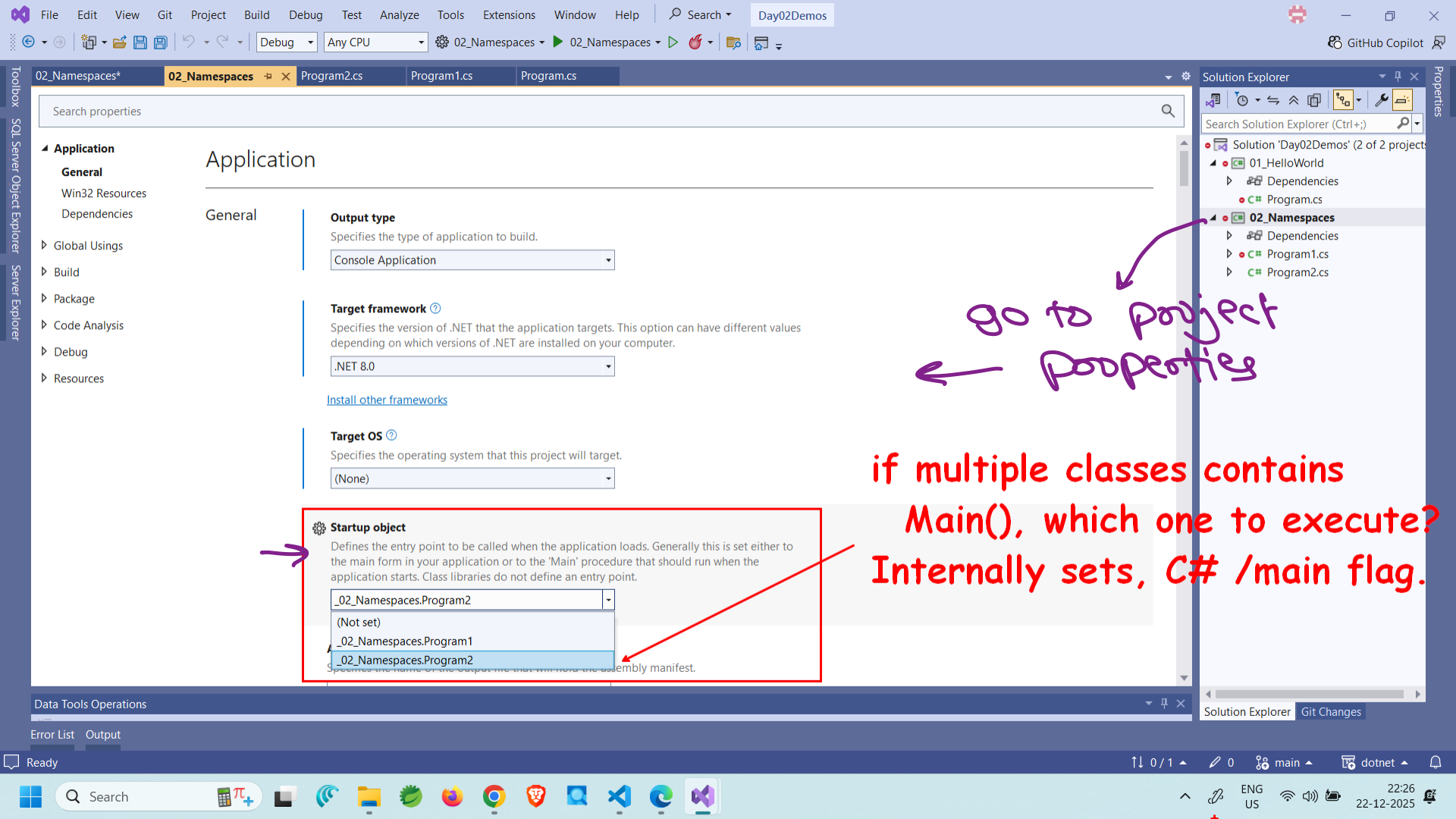
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SearchDay02Demos

GitHub Copilot

Program.csProgram2.cs02_Namespaces02_NamespacesProgram1.csProgram.cs

03_DataTypes03_DataTypes.ProgramMain(string[] args)

0 references

```
static void Main(string[] args)
{
    #region Type conversion using Parse()
    string str = "123";
    int num1 = int.Parse(str);
    Console.WriteLine($"num1 = {num1}");
    // int is converted to Int32 struct
    // Parse() is static method in struct
    // throw System.FormatException if input string is invalid
    #endregion

    #region Type conversion using Convert class
    int num2 = Convert.ToInt32(str);
    Console.WriteLine($"num2 = {num2}");
    // Convert class has plenty of static methods for converting
    // any type to other type (commonly used for primitive type conv)
    // throw System.FormatException if input string is invalid
    #endregion
}
```

104 %No issues foundLn: 17Ch: 13SPC CRLF

Data Tools Operations

Error ListOutput

Item(s) Saved

Solution Explorer

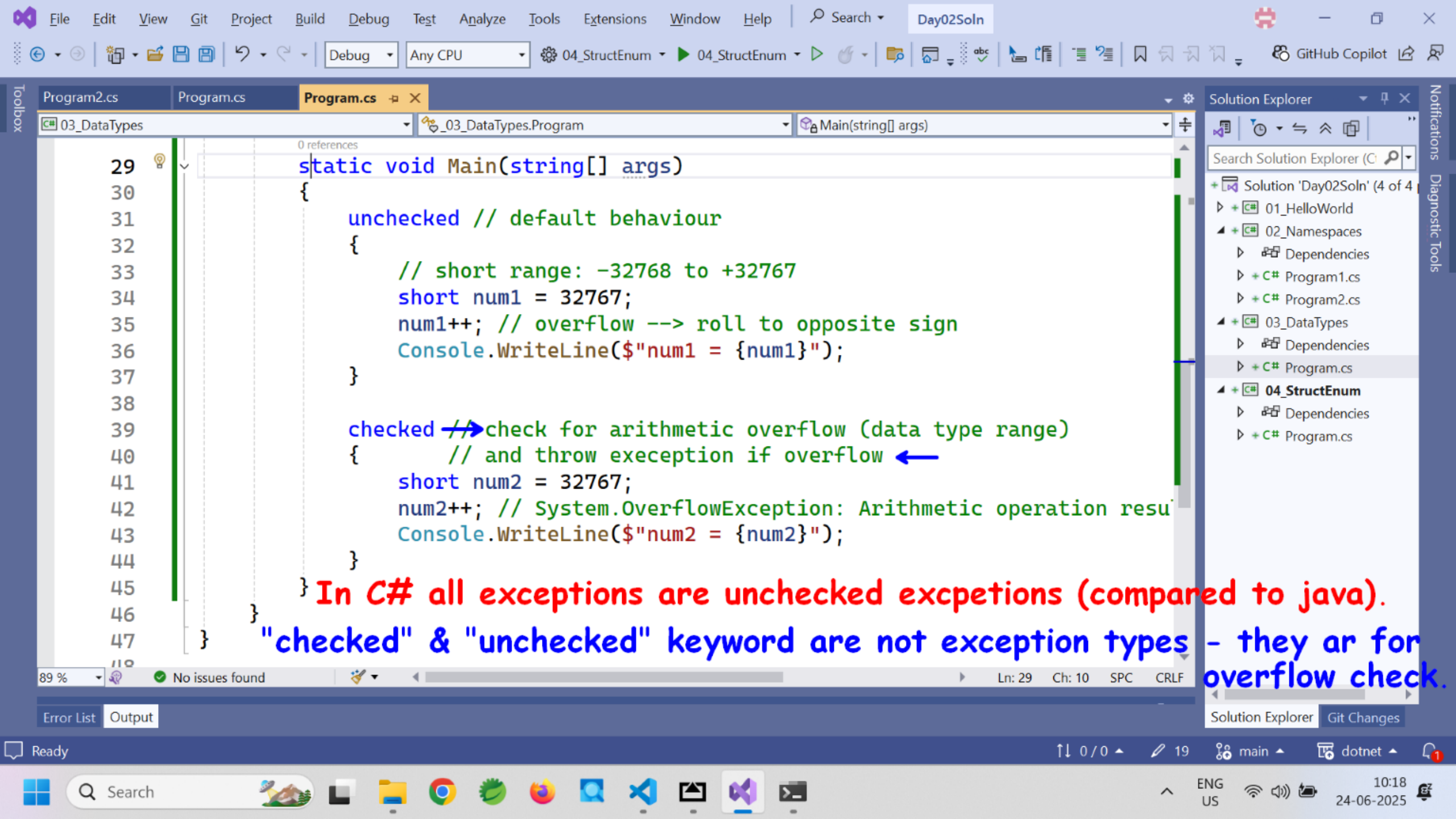
Search Solution Explorer (Ctrl+;)

Solution 'Day02Demos' (3 of 3 projects)

- 01_HelloWorld
 - Dependencies
- 02_Namespaces
 - Dependencies
- 03_DataTypes
 - Dependencies

SQL Server Object ExplorerServer Explorer

11:4722-12-2025



0 references

```
29 static void Main(string[] args)
30 {
31     unchecked // default behaviour
32     {
33         // short range: -32768 to +32767
34         short num1 = 32767;
35         num1++; // overflow --> roll to opposite sign
36         Console.WriteLine($"num1 = {num1}");
37     }
38
39     checked // check for arithmetic overflow (data type range)
40     {
41         // and throw exception if overflow
42         short num2 = 32767;
43         num2++; // System.OverflowException: Arithmetic operation result
44         Console.WriteLine($"num2 = {num2}");
45     }
46 }
47 }
```

In C# all exceptions are unchecked exceptions (compared to java).

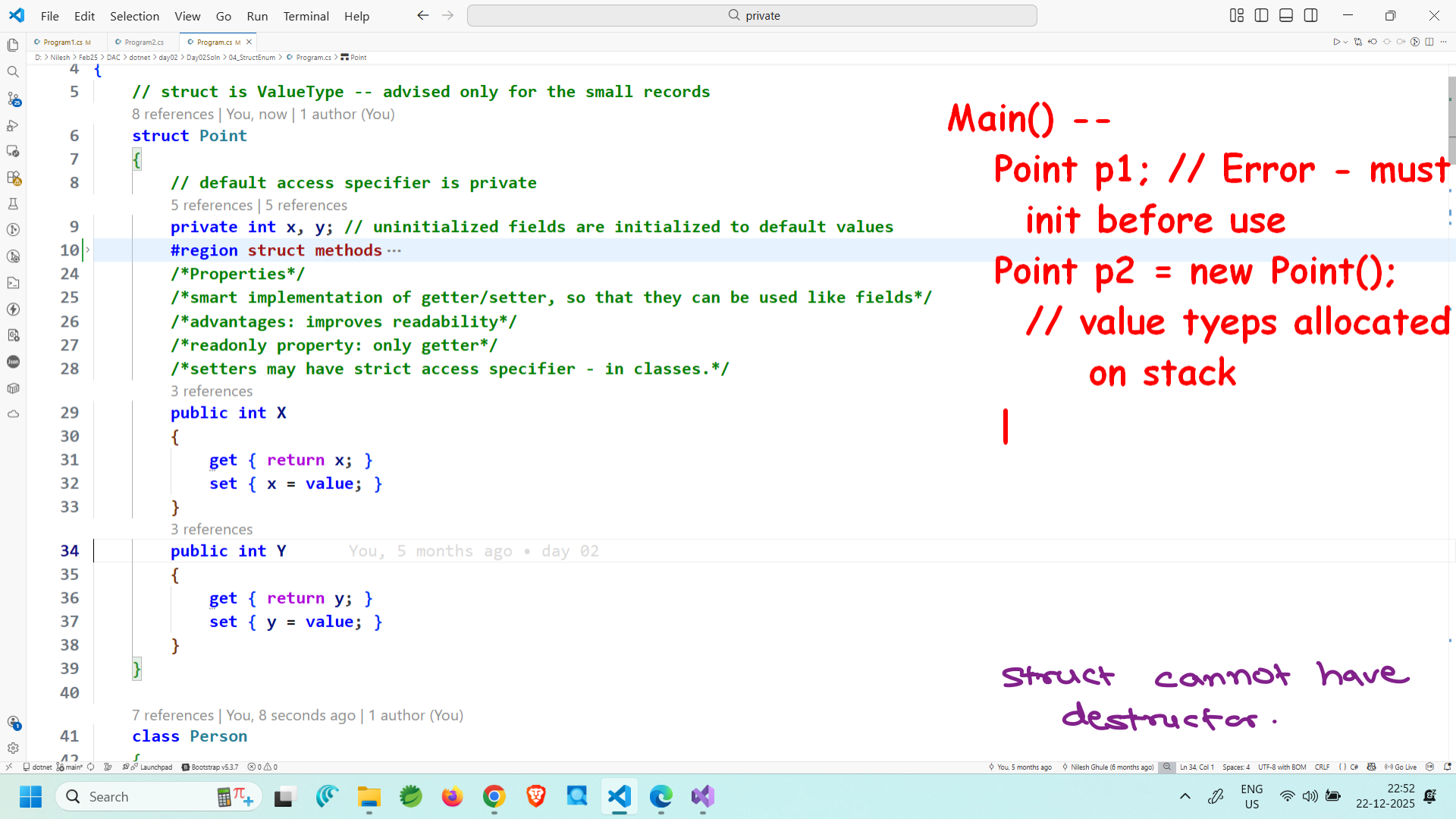
"checked" & "unchecked" keyword are not exception types - they are for overflow check.

Solution Explorer

Search Solution Explorer (C#)

- + Solution 'Day02Soln' (4 of 4)
 - + C# 01_HelloWorld
 - + C# 02_Namespaces
 - Dependencies
 - + C# Program1.cs
 - + C# Program2.cs
 - + C# 03_DataTypes
 - Dependencies
 - + C# Program.cs
 - + C# 04_StructEnum
 - Dependencies
 - + C# Program.cs

Solution Explorer | Git Changes

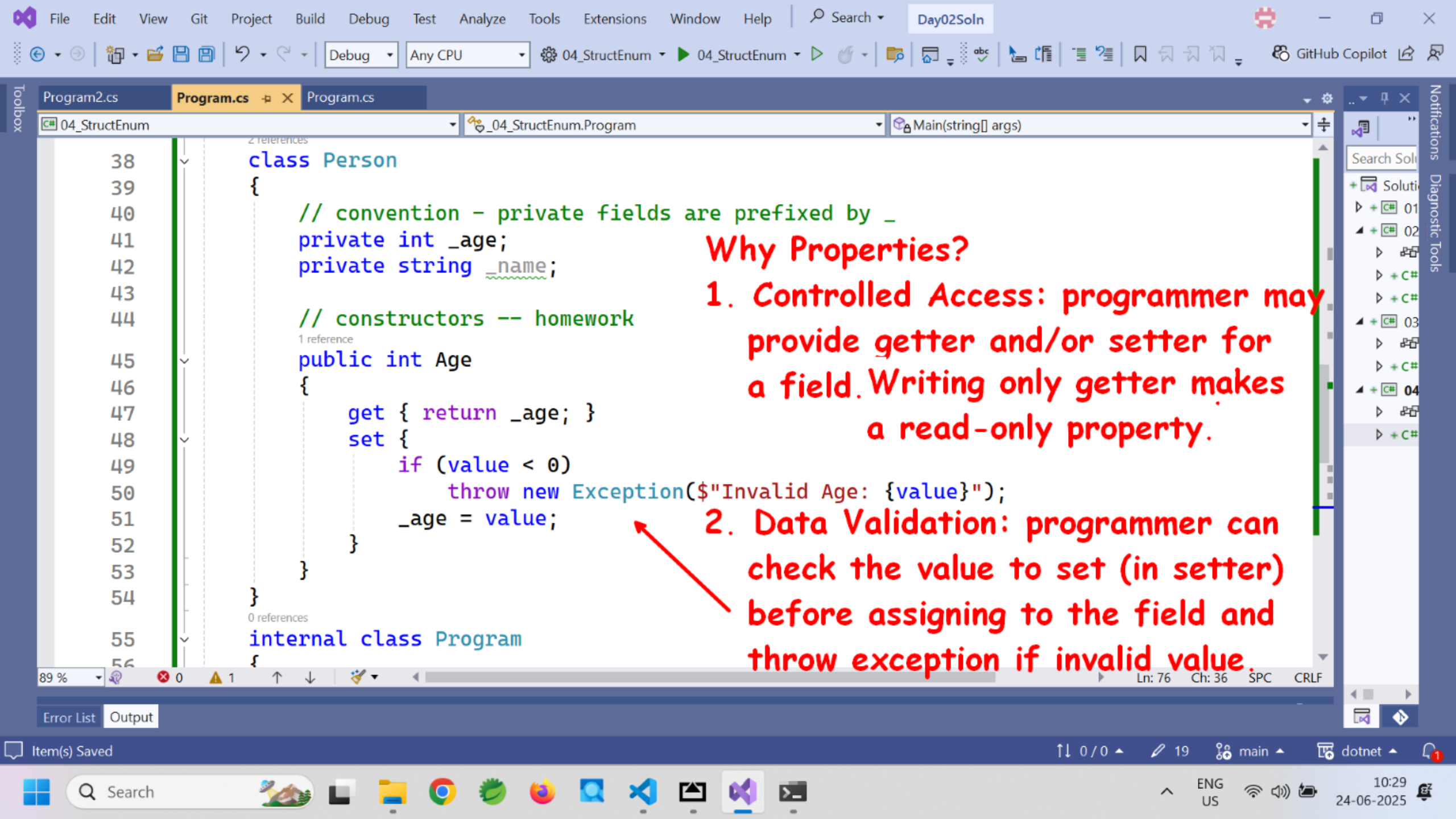


Main() --

Point p1; // Error - must
init before use

Point p2 = new Point();
// value types allocated
on stack

struct cannot have
destructor.



```
class Person
```

```
{
```

```
// convention - private fields are prefixed by _
```

```
private int _age;
```

```
private string _name;
```

```
// constructors -- homework
```

```
1 reference
```

```
public int Age
```

```
{
```

```
    get { return _age; }
```

```
    set {
```

```
        if (value < 0)
```

```
            throw new Exception($"Invalid Age: {value}");
```

```
        _age = value;
```

```
    }
```

```
}
```

```
0 references
```

```
internal class Program
```

```
{
```

Why Properties?

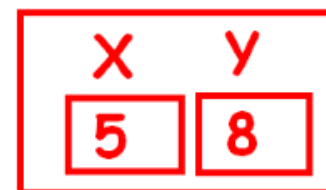
1. **Controlled Access:** programmer may provide getter and/or setter for a field. Writing only getter makes a read-only property.

2. **Data Validation:** programmer can check the value to set (in setter) before assigning to the field and throw exception if invalid value.

Stack

Heap

p3



Point obj

```
90  #region Class and Properties
```

```
91 Person pr1 = new Person();
```

```
92 //pr1.Age = -5; // System.Exception: Invalid Age: -5
```

```
93 pr1.Age = 5;           // okay -- classic property
```

```
94 pr1.Name = "John"; // okay -- auto-implemented property
```

```
95 pr1.Display();
```

```
96 #endregion
```

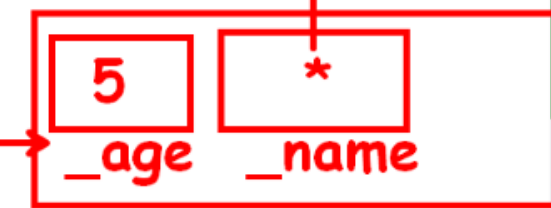
```
98  #region Object Initializer
```

```
99 // Object can be initialized while allocating it - using properties.
```

String obj

"John"

pr1



Person obj

Program1.cs M

Program2.cs

Program.cs M X

D:\Nilesh> Feb25 > DAC > dotnet > day02 > Day02Soln > 04_StructNum > Program.cs > Person

41

class Person

42

{

43

2 references

private int _age; // convention - private fields are prefixed by _

3 references

44 >

public Person()

You, 5 months ago • day 02 ...

3 references

48

public int Age

49

{

50

get { return _age; }

51

set

52

{

53

if (value < 0)

54

throw new Exception(\$"Invalid Age: {value}");

55

_age = value;

56

}

57

}

58

59

// auto-implemented property = field + basic getter/setter (no validation)

60

// field is created internally

61

// also minimal getter/setter created internally

4 references

62

public string Name

63

{

64

get;

65

set;

66

}

3 references

67

public void Display()

68

{

69

Console.WriteLine(\$"Person: Name={this.Name}, Age={this.Age}");

70

}

Person p1 = new Person();
// obj created on heap

p2 = new Person() {
Name="James", Age=65
};
p2.Display();

p3 = new Person() { Age=40 };
p3.Display();

obj init

conventional prop

auto-implemented prop

// collection initializer ...

list1=new List<string>() { "S1", "S2", "S3" };

dotnet main

Launchpad

Scotstrap v5.3.7

0 0

You, 5 months ago

Nilesh Ghule (6 months ago)

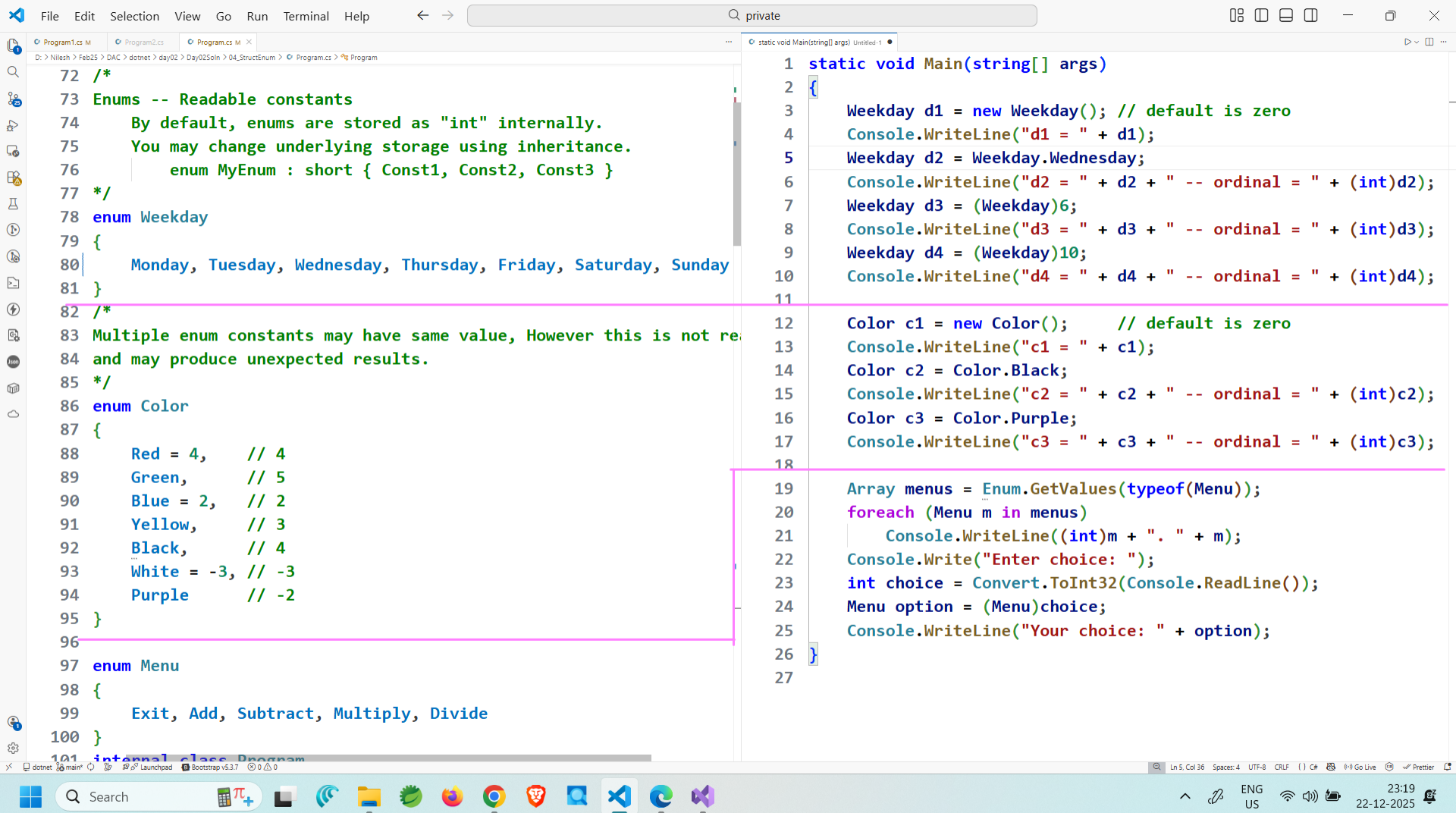
Ln 44, Col 1

Spaces: 4

UTF-8 with BOM

CRLF

Go Live



```
90 static void Main1(string[] args) ...
106 static void Main2(string[] args) ...
132 static void Main(string[] args)
133 {
134     Weekday d1 = new Weekday(); // def
135     Console.WriteLine("d1 = " + d1);
136
137     Weekday d2 = Weekday.Wednesday;
138     Console.WriteLine("d2 = " + d2 + "
139
140     Weekday d3 = (Weekday)6;
141     Console.WriteLine("d3 = " + d3 + "
142
143     Weekday d4 = (Weekday)10;
144     Console.WriteLine("d4 = " + d4 + "
145 }
146 }
147 }
```

89 % 0 1

Error List Output

Build succeeded

Microsoft Visual Studio Debug

```
d1 = Monday
d2 = Wednesday -- ordinal = 2
d3 = Sunday -- ordinal = 6
d4 = 10 -- ordinal = 10
```

```
D:\Nilesh\Feb25\DAC\dotnet\day02\Day02Soln\04_StructEnum\bin\Debug\net8.0\04_StructEnum.exe (process 27236) exited with code 0 (0x0).
Press any key to close this window . . .
```

FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchDay02Soln

04_StructEnum04_StructEnum

abc

GitHub Copilot

Program.csProgram2.csProgram.csProgram.cs

05_Boxing05_Boxing.ProgramMain(string[] args)

```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
internal class Program
{
    static void Main(string[] args)
    {
        int num1 = 123;
        object obj = num1;
        // up-casting
        // int (value type) --> object (reference type) = Boxing
        int num2 = (int)obj;
        // down-casting
        // object (reference type) --> int (value type) = Unboxing
    }
}
```

Stack

Heap

num1

123

obj

1000

123

num2

123

object

123

boxing

unboxing

Boxing - allcates new obj on heap and copy data

Unboxing - copies data from heap to var

Boxing/Unboxing .

- Needs more memory

- Needs time (copy)

89 %No issues foundLn: 18Ch: 2SPCCRLF

Error ListOutput

Solution Explorer

Search Solution Explore

Solution 'Day02Soln' (5 o

01_HelloWorld

02_Namespaces

Dependencies

Program1.cs

Program2.cs

03_DataTypes

Dependencies

Program.cs

04_StructEnum

Dependencies

Program.cs

05_Boxing

Dependencies

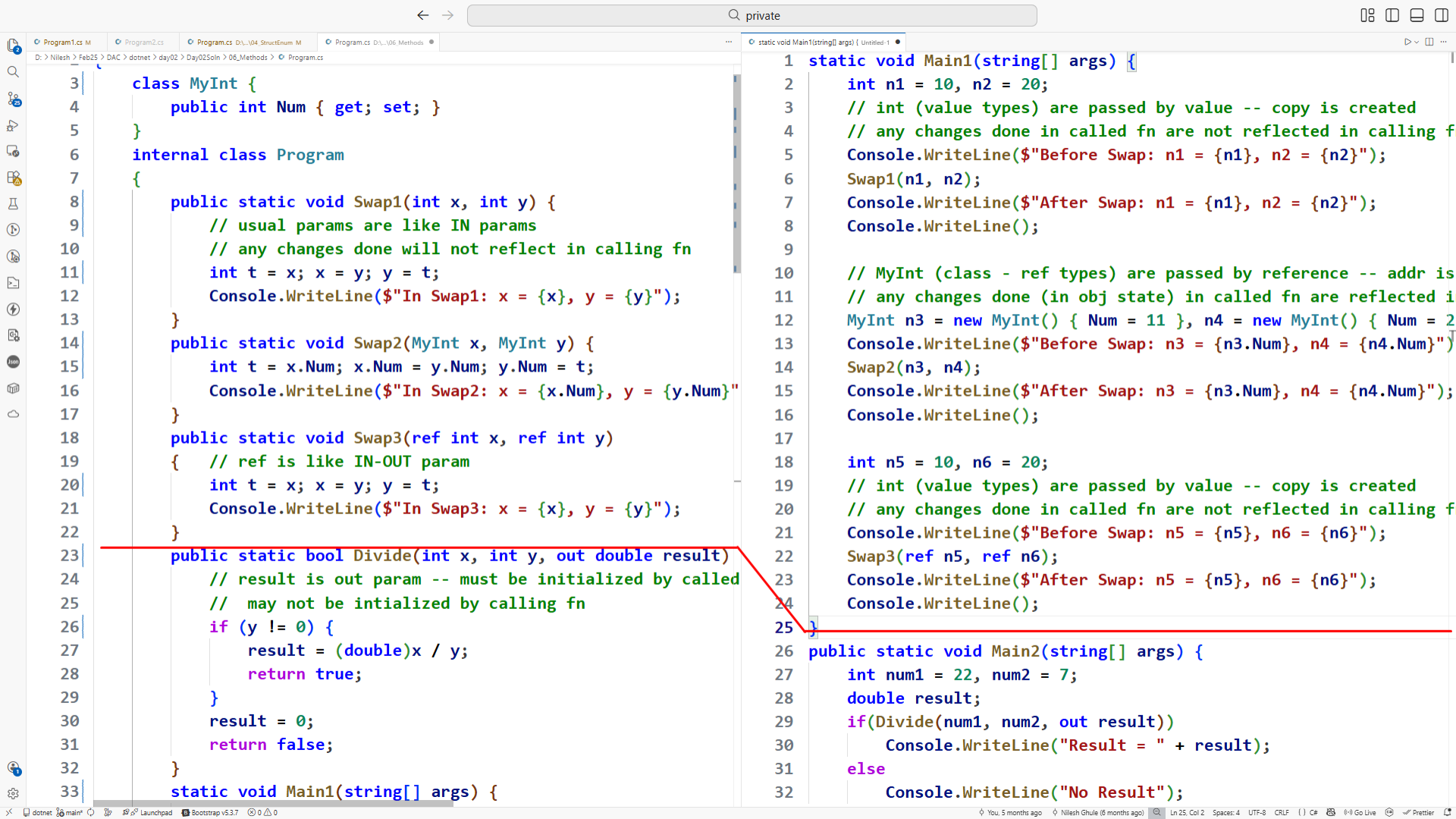
Program.cs

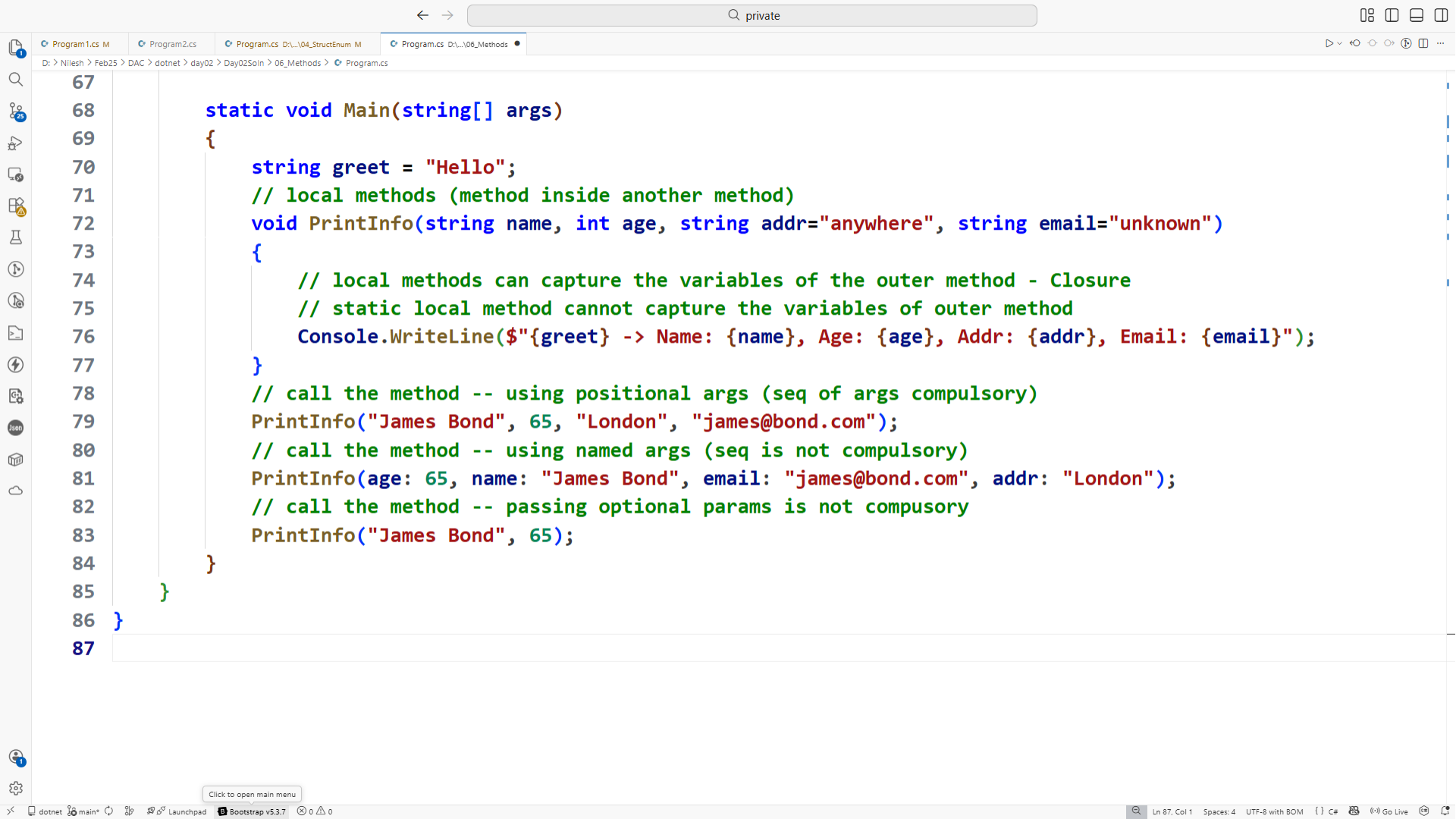
Solution Expl...Git Changes

Item(s) Saved

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11:2024-06-2025





Program1.cs M Program2.cs Program.cs D:_04_StructEnum M Program.cs D:_06_Methods

D: > Niles > Feb25 > DAC > dotnet > day02 > Day02Soln > 06_Methods > Program.cs

```
67
68     static void Main(string[] args)
69     {
70         string greet = "Hello";
71         // local methods (method inside another method)
72         void PrintInfo(string name, int age, string addr="anywhere", string email="unknown")
73         {
74             // local methods can capture the variables of the outer method - Closure
75             // static local method cannot capture the variables of outer method
76             Console.WriteLine($"{greet} -> Name: {name}, Age: {age}, Addr: {addr}, Email: {email}");
77         }
78         // call the method -- using positional args (seq of args compulsory)
79         PrintInfo("James Bond", 65, "London", "james@bond.com");
80         // call the method -- using named args (seq is not compulsory)
81         PrintInfo(age: 65, name: "James Bond", email: "james@bond.com", addr: "London");
82         // call the method -- passing optional params is not compulsory
83         PrintInfo("James Bond", 65);
84     }
85 }
86
87
```

Click to open main menu

```

9 internal class Person
10 {
11     public string Name { get; set; }
12     public int Age { get; set; }
13     public Person() : this("", 0)
14     {
15     }
16     public Person(string name, int age)
17     {
18         Name = name;
19         Age = age;
20     }
21     public virtual void Display()
22     {
23         Console.WriteLine($"Person: Name={this.Name}, Age={this
24

```

Person

Emp

```

3 static void Main(string[] args)
4 {
5     Person p1 = new Person("P1", 20);
6     p1.Display();
7     Console.WriteLine();
8
9     Emp e1 = new Emp("E1", 22, 1, 30000.0);
10    e1.Display();
11    Console.WriteLine();
12
13    Person p2 = new Emp("E2", 24, 2, 40000.0);
14    p2.Display();
15    Console.WriteLine();
16 }

```

Person

p1

P1 20

e1

E1 22

1 30000 Emp

p2

E2 24

2 40000 Emp

```

2 namespace _07_OopInheritance
3 {
4     class Emp : Person
5     {
6         public int Id { get; set; }
7         public double Salary { get; set; }
8
9         public Emp():this("", 0, 0, 0.0)
10        {
11        }
12        public Emp(string name, int age, int id, double salary)
13            :base(name, age)
14        {
15            this.Id = id;
16            this.Salary = salary;
17        }
18        public override void Display()
19        {
20            base.Display();
21            Console.WriteLine($"Emp: Id={this.Id}, Salary={this.Sal
22        }
23    }
24 }

```

Emp

- Id
- Salary
- CalcIncome()
- DoWork()

EmpUtil

- CalcTotalExp()

Labor $\rightarrow$ Total sal = hours * rate

Manager $\rightarrow$ CalcIncome()

- Salary + Bonus

(Mgrs won't get extra component).

Sales Manager & HR Manager.

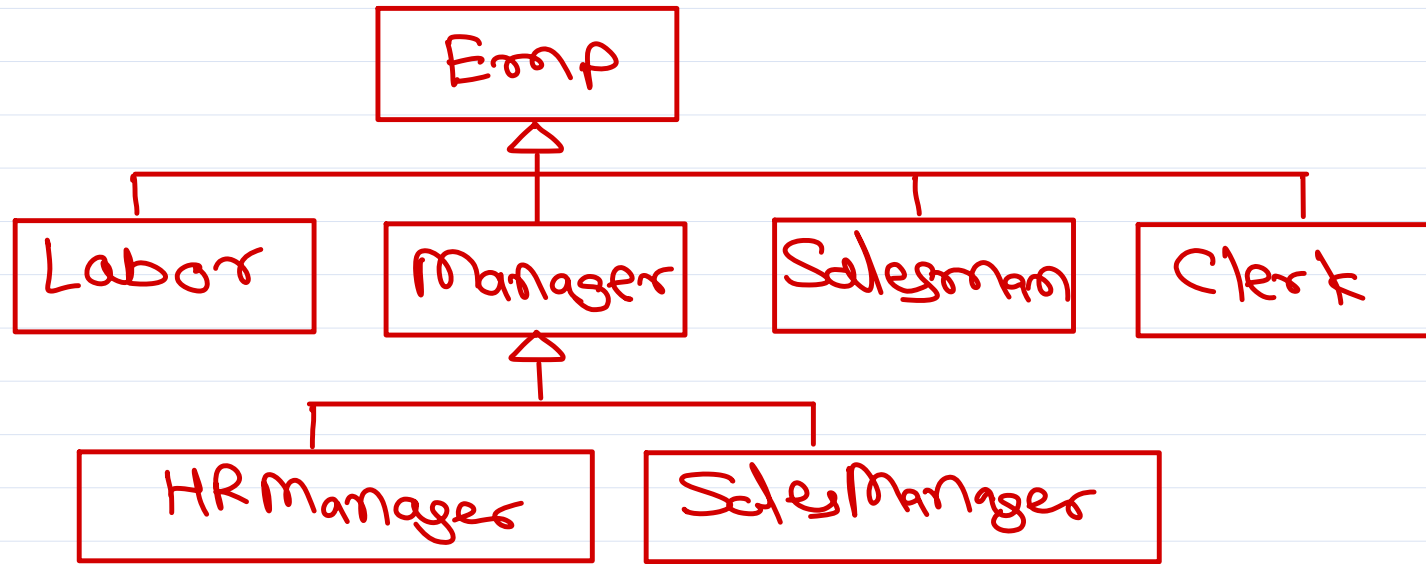
Salesman $\rightarrow$ CalcIncome()

- Salary + Commission

Clerks with no sub-types.



Inheritance



Object Oriented Analysis & Design (OOAD)
- Grady Booch.

* SOLID

* Design Patterns

abstract methods
(contract)

conceptual entity
(no objects)

classes

- fields + methods (reuse).
- virtual methods (polymorphism)
- objects

abstract classes

- fields + methods (reuse).
- virtual methods (polymorphism)



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SearchDay03_Soln

01_Struct01_Struct

GitHub Copilot

Program.csProgram.cs

02_OopInheritance02_OopInheritance.ProgramMain(string[] args)

59static void Main(string[] args)

60{

61Emp[] emps = new Emp[5];

62// array of 5 references created on heap

63// by default each ref is initialized to null

64}

65}

66}

67

0 references

emps

1000

Array

Length

5

_data

0null

1null

2null

3null

4null

89 %No issues foundLn: 59Ch: 6SPCCRLF

Error ListOutput

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'Day03_Soln' (2 of 2 projects)

01_Struct

02_OopInheritance

Dependencies

Program.cs

Item(s) Saved

0/011maindotnet

ENG US08:4125-06-2025

Object Slicing (Java/C#) -

When derived class object is assigned to base class ref,
we can only access base class members through that ref.

```
class Person {  
    public string name;  
    // ...  
}
```

```
class Student:Person {  
    public int marks;  
    // ...  
}
```

```
Person p = new Student();  
cw(p.name); // okay  
cw(p.marks); // error
```

Exception - The overridden virtual
members are accessed based on
type of object (not the ref).

```
Student s = (Student) p;  
cw(s.marks); // okay
```



Thank you!

Nilesh Ghule <nilesh@sunbeaminfo.com>

